

Body Control Module (BCM) - High-Level Design

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This is a SYNTHETIC document created for academic evaluation of a RAG assistant. It contains no real company or customer data.

1. Introduction and Scope

This document describes the high-level design of the Body Control Module (BCM). The BCM controls the central door locks, the power windows and the exterior lighting of the vehicle.

The BCM software is built on the AUTOSAR Classic Platform, release R21-11. The application layer contains five software components: DoorLockManager, WindowControl, LightingController, PowerModeManager and BcmDiagnostics.

Out of scope for this release: wipers, climate control (HVAC) and passive keyless entry.

2. Architecture Overview

The software is organised in layers: the application layer with five software components, the Runtime Environment (RTE), the Basic Software (BSW) services (Com, NvM, Dem, Dcm), the ECU abstraction layer and the microcontroller abstraction layer (MCAL).

The BCM is connected to the body CAN bus at 500 kbit/s. The window switches are connected over LIN at 19.2 kbit/s.

Component	Type	Runnable period	Owner team
DoorLockManager	Application SWC	10 ms	Body Team A
WindowControl	Application SWC	20 ms	Body Team B
LightingController	Application SWC	50 ms	Lighting Team
PowerModeManager	Service SWC	10 ms	Platform Team
BcmDiagnostics	Application SWC	100 ms	Diagnostics Team

3.1 DoorLockManager

DoorLockManager locks and unlocks the four doors. It handles central-locking requests, automatic locking while driving and crash unlock.

Its runnable DoorLock_Run10ms executes every 10 ms. The lock actuators are driven through a Dio channel group.

Crash unlock: when the signal CrashDetected is set to 1 by the airbag control unit, DoorLockManager unlocks all doors within 100 ms.

Auto-lock: when VehicleSpeed is above 10 km/h, all doors are locked automatically.

The last lock state is stored in the NvM block NvM_DoorLockState so it survives a power cycle.

3.2 WindowControl

WindowControl drives the four window motors. It supports manual up/down and one-touch up/down. Its runnable executes every 20 ms.

Anti-pinch: during one-touch close, an obstruction is detected from the motor current. On detection the motor is reversed for 200 ms.

The maximum one-touch travel time is 4 seconds.

Window operation is disabled when the battery voltage is below 9 V, to protect the battery.

The window switch inputs arrive over LIN from the door switch pods.

3.3 LightingController

LightingController controls the low beam, high beam, daytime running lights (DRL) and the turn indicators. Its runnable executes every 50 ms.

Automatic headlights: the low beam is switched on when the ambient light is below 400 lux for 5 seconds, and switched off when it is above 800 lux for 5 seconds. The gap between 400 and 800 lux is a hysteresis.

The turn indicator flash frequency is 1.5 Hz (90 flashes per minute). Hazard lights override the turn indicators.

Bulb failure is detected by current monitoring. When a bulb failure is detected, the indicator flash rate doubles to 3 Hz to warn the driver.

3.4 PowerModeManager and BcmDiagnostics

PowerModeManager controls the ECU power state. There are three BCM power modes: SLEEP, STANDBY and RUN.

Wake-up sources are a CAN wake-up, a door switch and the ignition signal.

After ignition off, the BCM stays in RUN for 30 seconds and then moves to STANDBY. After 10 minutes without activity, STANDBY moves to SLEEP. In SLEEP the current draw must stay below 100 microamps.

BcmDiagnostics implements UDS over CAN and reports faults to the Diagnostic Event Manager (Dem). It runs every 100 ms.

4. Interfaces and Ports

The table lists the main ports of the BCM software components.

Port name	Interface type	Provider	Consumer
Pp_LockState	Sender-Receiver	DoorLockManager	BcmDiagnostics
Rp_VehicleSpeed	Sender-Receiver	Com (CAN)	DoorLockManager
Rp_CrashDetected	Sender-Receiver	Com (CAN)	DoorLockManager
Pp_WindowPosition	Sender-Receiver	WindowControl	BcmDiagnostics
Rp_BatteryVoltage	Sender-Receiver	IoHwAb (ADC)	WindowControl
Rp_AmbientLux	Sender-Receiver	IoHwAb (ADC)	LightingController
Pp_BcmMode	Mode-Switch	PowerModeManager	All other SWCs
Cs_NvmWrite	Client-Server	NvM	DoorLockManager

5. CAN Signals and Messages

All messages below are on the body CAN bus at 500 kbit/s.

Signal	Message	CAN ID	Direction	Cycle
VehicleSpeed	ESC_Status	0x1A0	Rx	20 ms
CrashDetected	ACU_Crash	0x0C5	Rx	Event-triggered
IgnitionState	IGN_Status	0x120	Rx	50 ms
DoorLockStatus	BCM_Body1	0x3B0	Tx	100 ms
WindowPos_FL	BCM_Window	0x3B8	Tx	100 ms
HeadlampState	BCM_Light	0x3C0	Tx	200 ms

6. Functional Flow: Central Locking

The central-locking flow has six steps:

1. A lock request arrives over CAN in the signal LockRequest.
2. PowerModeManager moves the BCM to RUN mode if it is not already in RUN.
3. DoorLockManager checks that all doors are closed using the door switches.
4. If any door is open, the lock request is rejected and a warning chime is requested.
5. Otherwise the lock actuators are driven with a pulse of 400 ms.
6. DoorLockStatus is updated on CAN and the new state is written to NvM.

7. Functional Flow: Automatic Headlights

The automatic headlight flow works as follows:

1. The ambient light value (Rp_AmbientLux) is read every 50 ms.
2. The value is filtered with a moving average over 8 samples.
3. If the filtered value stays below 400 lux for 5 seconds, the low beam is switched on.
4. If the filtered value stays above 800 lux for 5 seconds, the low beam is switched off.
5. A manual switch position always overrides the automatic function.

Fail-safe: if the light sensor is faulty, the low beam is switched on by default.

8. Dependencies and BSW Configuration

The table shows which Basic Software modules each software component depends on.

Software component	Depends on BSW modules
DoorLockManager	Com, NvM, Dio
WindowControl	Com, LinIf, Adc (via IoHwAb)
LightingController	Pwm, Adc, Com
PowerModeManager	EcuM, BswM, ComM
BcmDiagnostics	Dcm, Dem, Com

The operating system has four cyclic tasks: Task_10ms (highest priority), Task_20ms, Task_50ms and Task_100ms (lowest priority).

The NvM block NvM_DoorLockState has a size of 4 bytes. Estimated flash usage is 1.2 MB of 2 MB, and estimated RAM usage is 70 KB of 128 KB.

9. Diagnostics and Error Handling

The BCM supports these UDS services: 0x10 DiagnosticSessionControl, 0x22 ReadDataByIdentifier, 0x19 ReadDTCInformation and 0x14 ClearDiagnosticInformation.

DTC	Description	Trigger
0xB10001	Door lock actuator open circuit	No actuator current during the 400 ms lock pulse
0xB10002	Window anti-pinch triggered	Anti-pinch reversal detected
0xB20001	Headlamp bulb failure	Bulb current below the expected range
0xC10001	Lost communication with ESC	No ESC_Status message received for 100 ms

Fault reaction: if VehicleSpeed is missing (DTC 0xC10001), the auto-lock function is disabled.

10. Revision History and Open Issues

Version	Date	Author	Change
0.1	2026-07-01	Architecture Team	Initial draft
0.5	2026-08-10	Architecture Team	Added window and lighting components
1.0	2026-09-15	Architecture Team	Approved baseline

Open issues:

OI-01: Confirm the anti-pinch force threshold with the supplier (owner: Body Team B).

OI-02: Decide whether remote window close should be supported.

OI-03: Define the method to measure the SLEEP current.